

Cybersecurity

Managing Permissions and Services

Lesson 4.3



Learning Objectives

By the end of today's class, you will be able to:

- 1 Inspect and set file permissions for sensitive files on the system.
- 2 Manage and monitor services on the system.
- 3 Create and assign users for services.



Welcome



Recap

Linux Access Controls

01

Audit passwords using `john`

02

Elevate privileges with `sudo` and `su`

03

Create and manage users and groups



Access **Controls** and Permissions



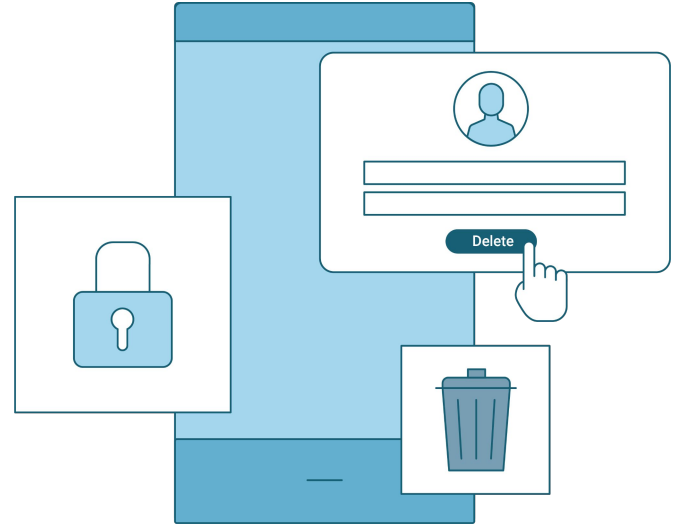
Like Google Docs, Linux has **access controls** that grant permission to access documents and files on a host.

Access Controls and Permissions

Access controls allow actions on a file to include:

- Read
- Write
- Execute
- Delete
- Rename

Permissions regulate who can take which action.

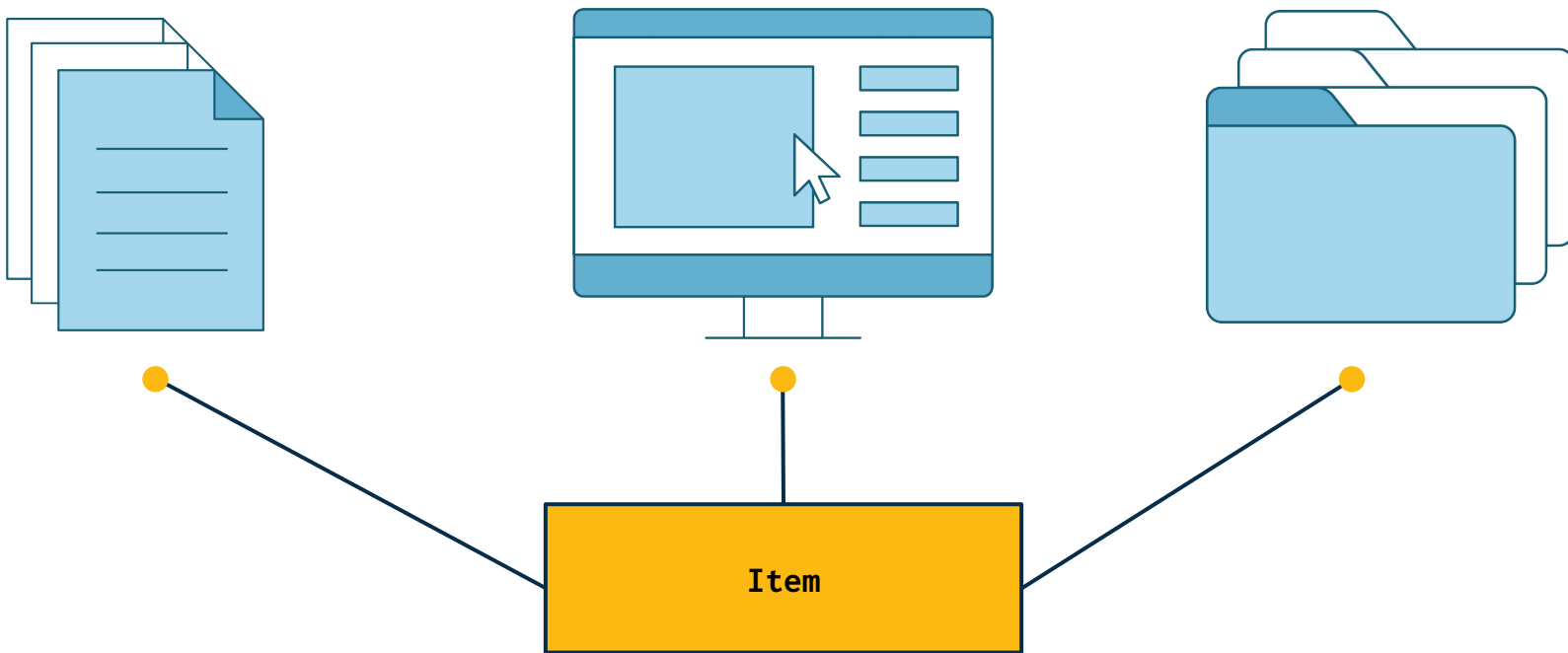




Managing Access Controls in Linux

Setting Permissions

Linux categorizes files, programs, and directories as **items**.





Managing Access Controls in Linux

Setting Permissions

Each item specifies access controls for the **owner**, the **group**, and **others**.

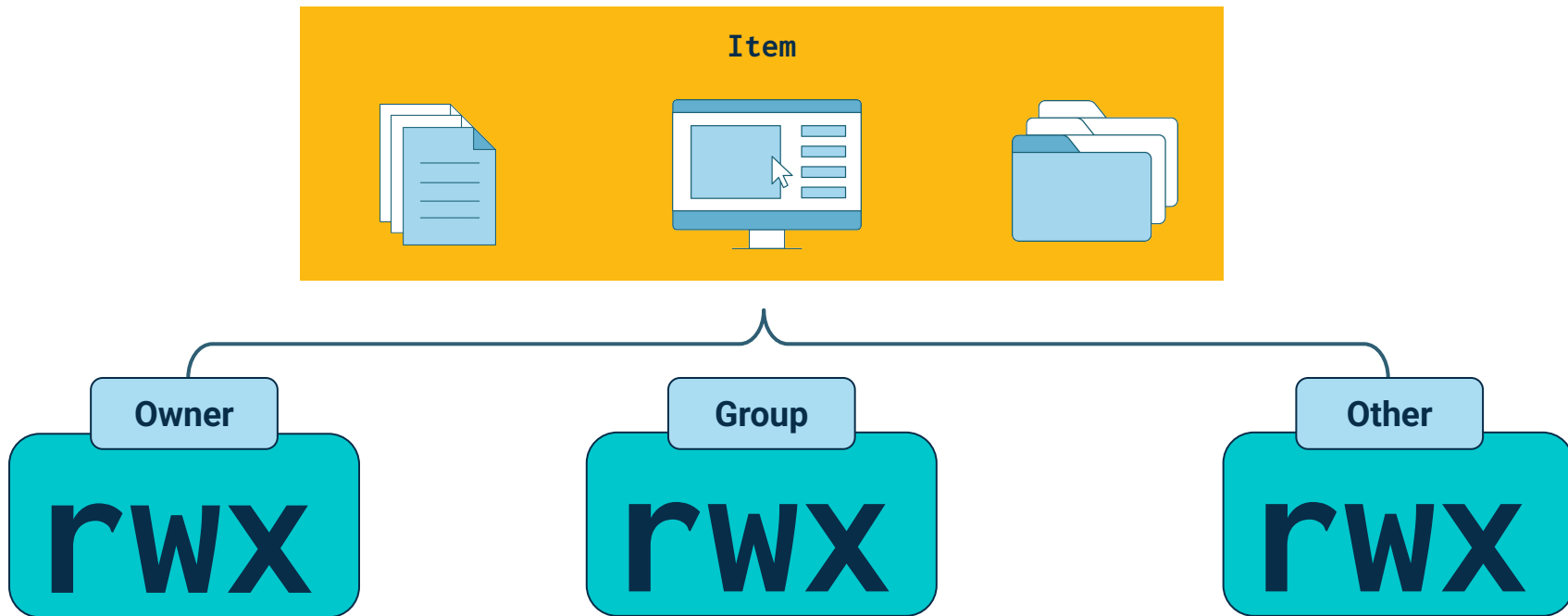




Managing Access Controls in Linux

Setting Permissions

Each item specifies access controls for the **owner**, the **group**, and **others**.



Access Controls and Permissions

Linux Filesystem Permissions

Owner

User that created the item.



Group

Primary group of the owner.



Other

Everyone who is **not** the owner or not in the group.



Access Controls and Permissions

Options for Owner, Group, and Other

Read

Determines if the item can be read or viewed.

r

Write

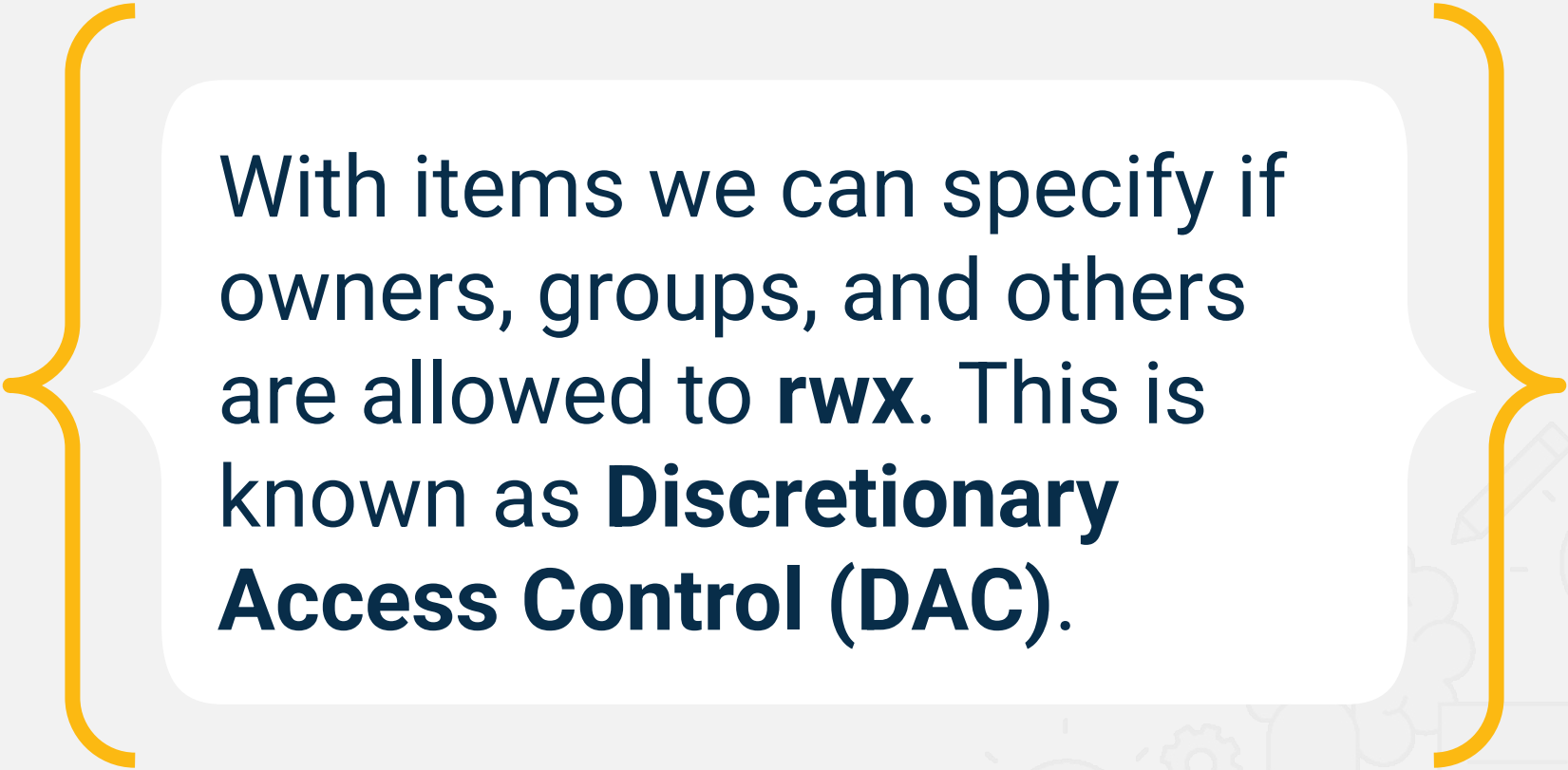
Determines if the item can be written to or modified.

w

Execute

Determines if the item can be executed.

x



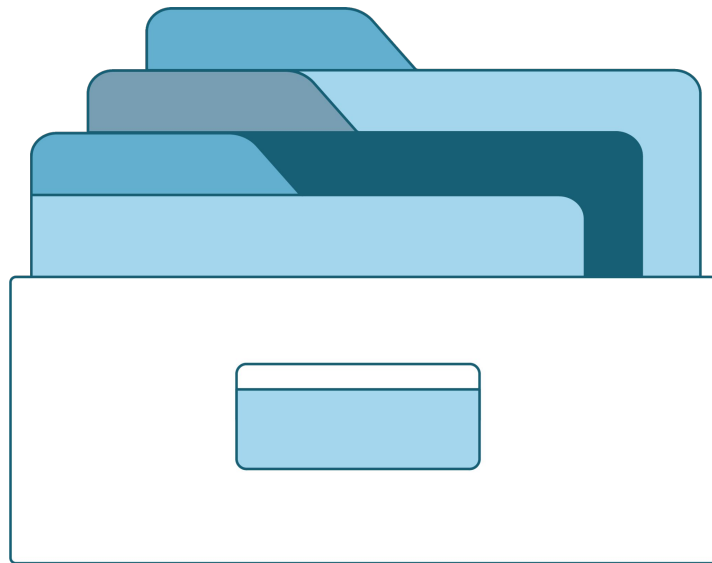
With items we can specify if owners, groups, and others are allowed to **rwX**. This is known as **Discretionary Access Control (DAC)**.

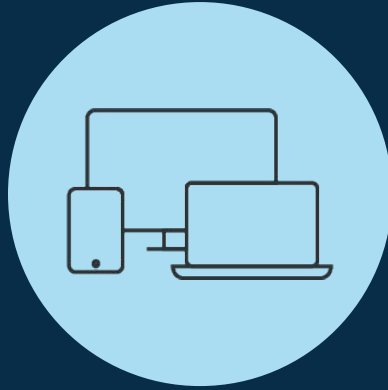




Demonstration Scenario

- We will create a file and a directory, and examine the default permissions.
- We will also need to change these permissions to deny particular users and groups access.
- Commands used:
 - `ls -l`
 - `chmod`
 - `chown`





Instructor Demonstration

Permissions

Inspecting File Permissions

-rw- r - - r - -

Item type (- for file, d for directory)

Inspecting File Permissions

- **rw-** r - - r - -

Owner permission (in this case, **read** and **write**)

Item type

Inspecting File Permissions

- rw- r - - r - -

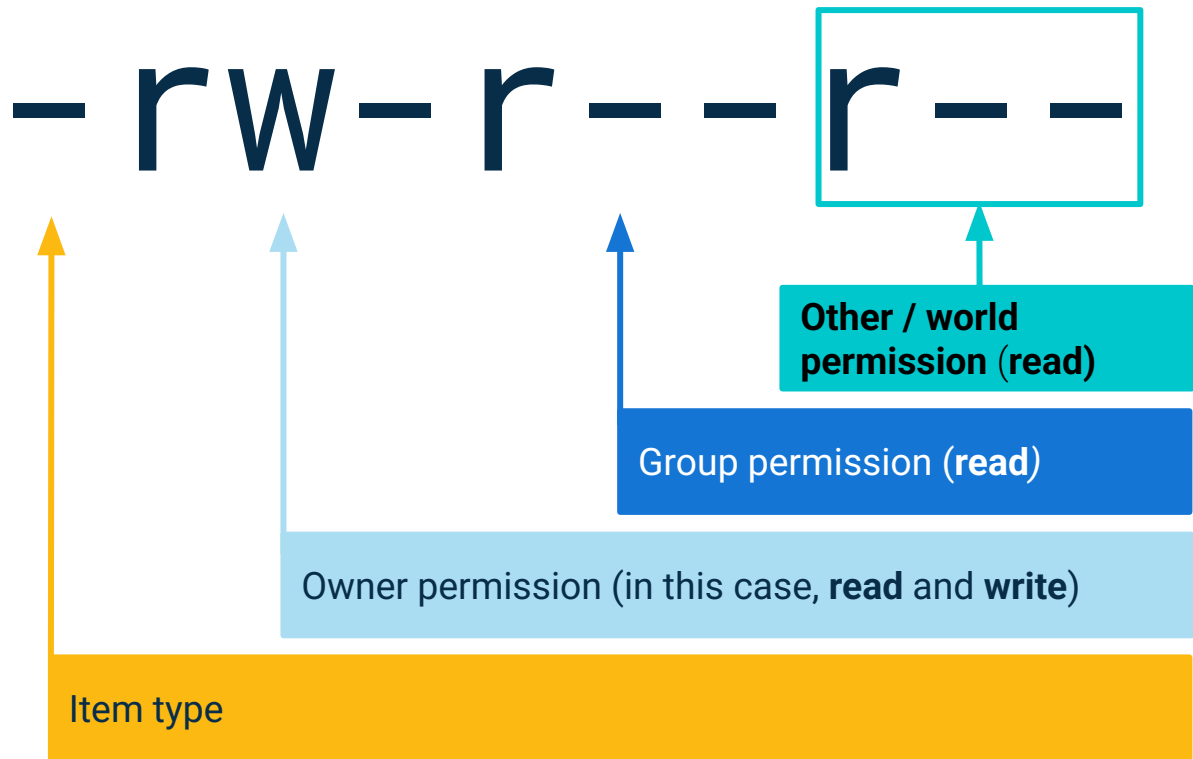


Group permission (read)

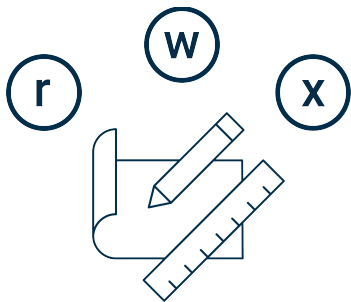
Owner permission (in this case, **read** and **write**)

Item type

Inspecting File Permissions



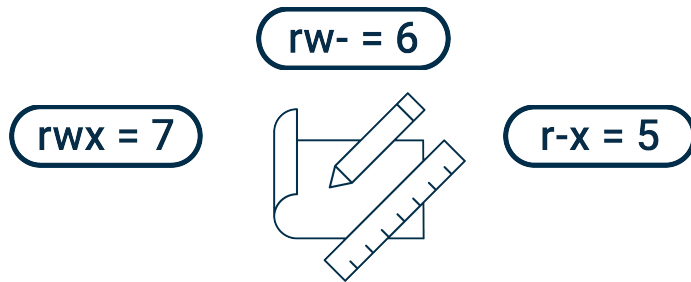
Changing Permissions



Symbolic Notation

Uses letters to set permissions

- r
- w
- x



Octal Notation

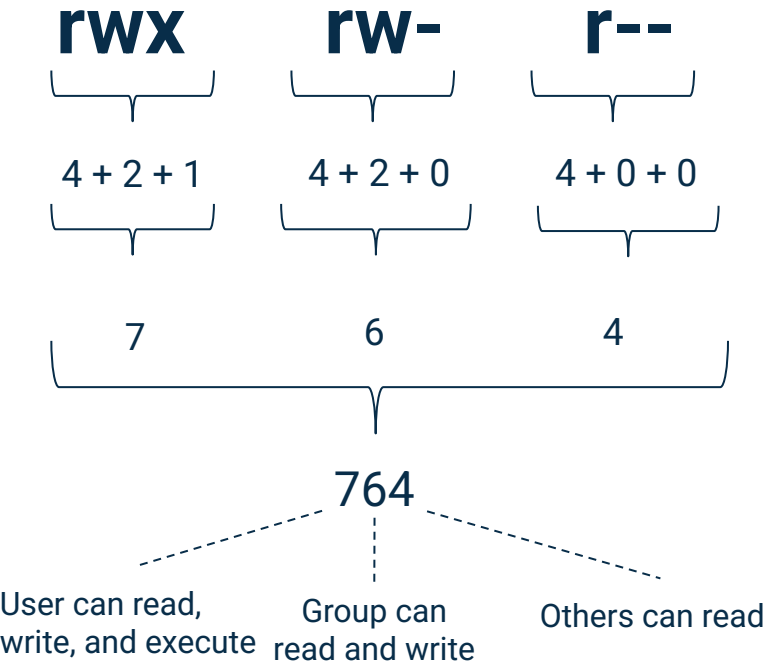
Uses base-8 numbers to set permissions

- rwx = 7
- rw- = 6
- r-x = 5

Changing File Permissions

Symbolic and Octal Notation

Octal Notation				
	4	2	1	
0	-	-	-	No permission
1	-	-	x	Only execute
2	-	w	-	Only write
3	-	w	x	Write and execute
4	r	-	-	Only read
5	r	-	x	Read and execute
6	r	w	-	Read and write
7	r	w	x	Read, write, and execute





Demo Summary

Let's Recap



`ls -l`



`chown user:group`



`chmod`



Symbolic & octal notation



Activity:

Access Controls and Permissions

In this activity, you will inspect and set file permissions on some of the most sensitive files on the Linux system.

- The senior administrator would like you to make sure a few important files and directories are secure by checking and modifying their permissions.

Suggested Time:
25 Minutes



Time's up!
Let's review



Questions?

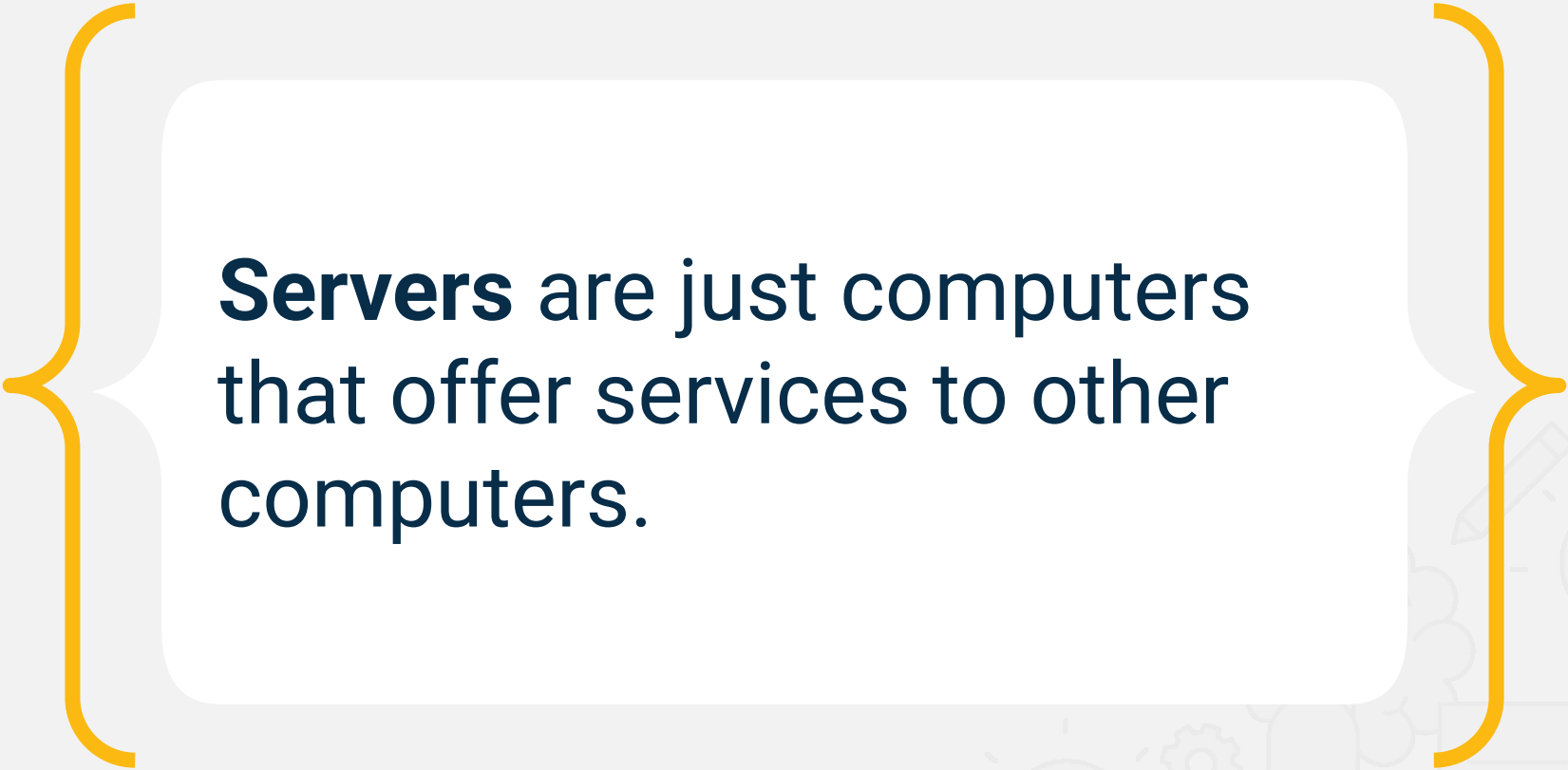




Break
15 mins



Managing Services



Servers are just computers
that offer services to other
computers.



Managing Services

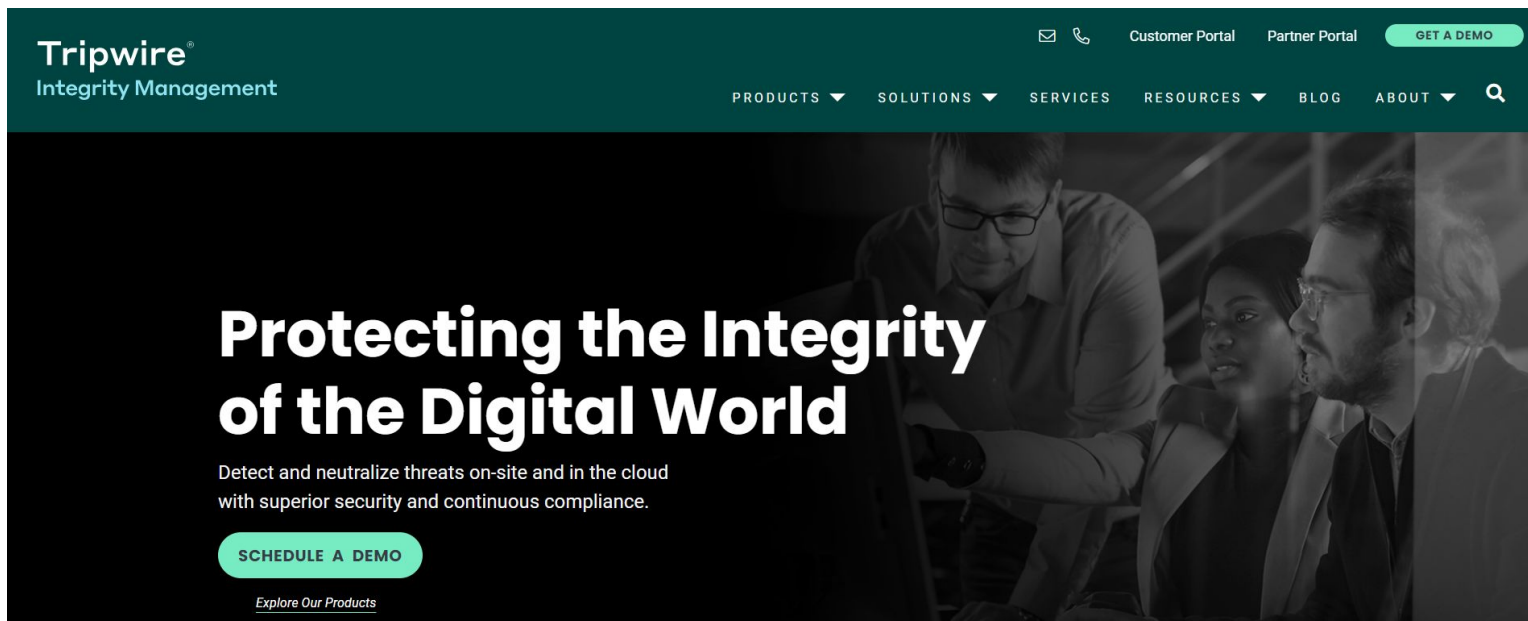
A **service** is a function or capability that a machine makes available to another.

For example, file-sharing services allow computers to send and receive data.



Managing Services

Some services, like Tripwire, are only run locally on the server and are not provided to other computers. These services are packages that can be installed and removed just like other programs.



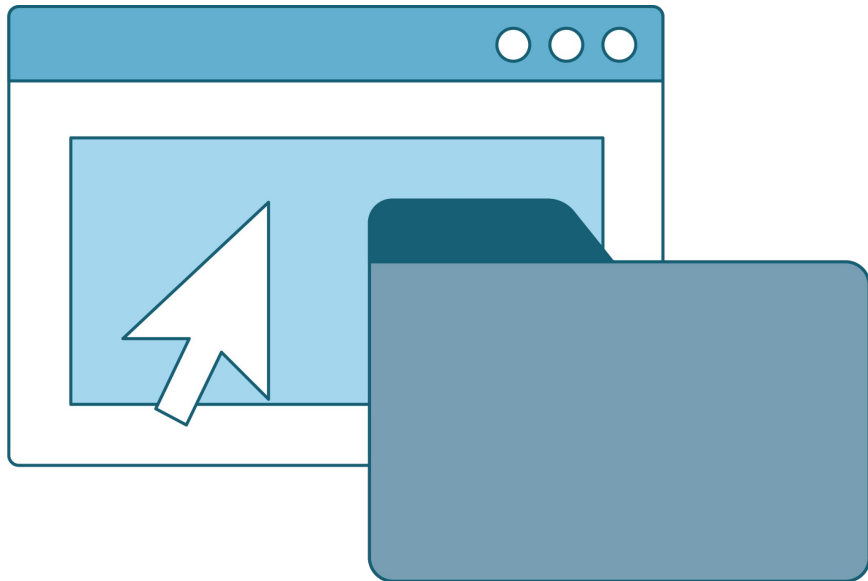


SMB

Server Message Block

Server Message Block (SMB):

- File sharing protocol
- Windows devices
- Open-source version is Samba
- Useful, but risky

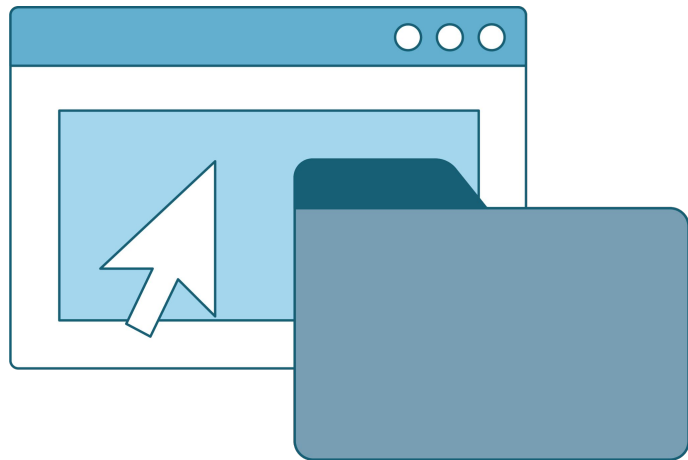


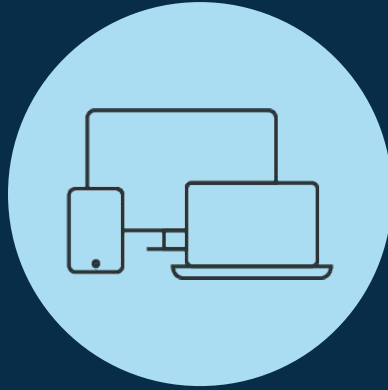


Demonstration Scenario

The server has already been **compromised**, and you will proceed by stopping the SMB service, and then uninstalling it from the system.

- Steps required:
 - a. List all running services.
 - b. Identify the Samba service in the list to verify that it's running, then stop it.
 - c. Ensure that Samba doesn't start when the machine is started up.
 - d. Ensure that Samba is no longer running.
 - e. Uninstall the Samba service completely.





Instructor Demonstration

Finding and Stopping **SMB**



Activity:

Managing Services

In this activity, you will continue auditing the services running on this server.

- Your senior administrator wants you to audit the services being run on the server and shut down old and unused services. They have provided a list of services to check.
- You will check for this list of services and shut down and remove any old service that you find.

Suggested Time:
25 Minutes



Time's up!
Let's review



Questions?





Service Users



Services should be run
by a user dedicated to
running that service.



Service Users

User ID

A service user usually has a system UID less than 1000 and cannot log in to use a shell.

UID

```
root:x:0:0:root:/root:/bin/bash
```

Service Users



Since service users aren't humans who need to log into and interact with the machine, it's best practice to ensure that users cannot log into an interactive shell using a service username.

- `ftp` user unable to login to a bash session
- Reduces attack surface



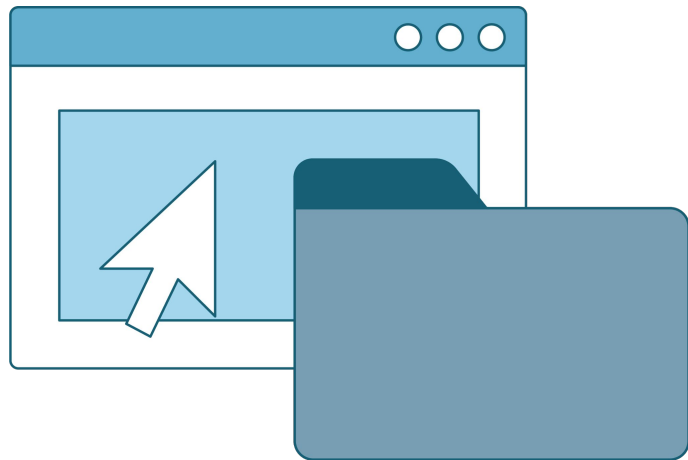
Demonstration Scenario

Setup, Removing, and Adding Service

Your senior administrator has asked you to follow up on uninstalling unused services.

You must now ensure the services' corresponding users have also been removed from the system.

Previously, you disabled `vsftpd`, but its service user, `ftp`, still exists.



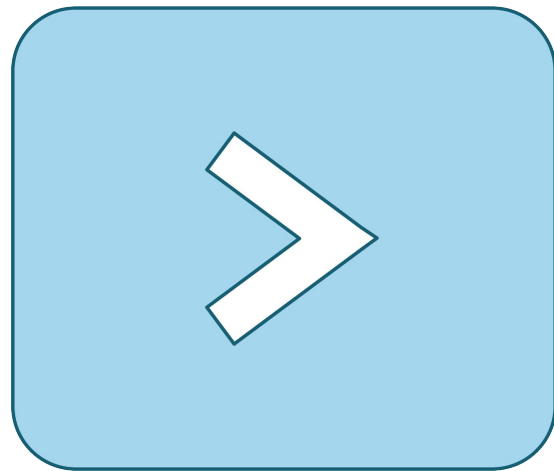


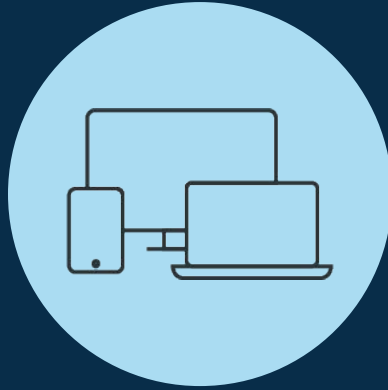
Demonstration Scenario

Splunk Service Account

Additionally, your senior administrator plans to install Splunk.

They've informed you that they'll handle the installation, but have requested that you create a **service user**.





Instructor Demonstration

Set Up, Removing, and Adding **Service** Users



Activity:

Service Users

In this activity, you will continue with auditing the same server for your senior administrator.

The senior administrator would like to remove any old service users from the system and create a new user that will be dedicated to running the Tripwire program.

Suggested Time:
25 Minutes



Time's up!
Let's review



Questions?





Recap

Managing Permissions and Services

01

Set file permissions for sensitive files on the system.

02

Managed and monitored services on the system, removing unused.

03

Assigned users for services.



Next class

Create backups and restore data



Thank you